

Book 2: Chapter 3 – Distance Metrics

Machine Learning for Asset Managers (Marcos M. López de Prado)

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Chapter 3 Overview: Why Distance Metrics Matter

The Problem with Traditional Correlation

- **Linear Constraint:** Pearson's ρ only measures *linear* dependence, missing non-linear relationships.
- **Outlier Sensitivity:** Single extreme outliers heavily distort correlation estimates.
- **Non-Metric Space:** Raw correlation $\rho \in [-1, 1]$ is not a valid distance metric (violates triangle inequality).

The Information-Theoretic Solution

- **Metric Spaces:** Converting correlation into true metric distances ($d_\rho, d_{|\rho|}$).
- **Entropy & Mutual Information:** Measuring general (linear + non-linear) dependency.
- **Variation of Information (VI):** A bounded, robust metric for clustering and feature selection.

Topic 3.1: Correlation-Based Distance Metrics

Mathematical Formulations

To use correlation in unsupervised learning (e.g., clustering), we must transform $\rho(X, Y)$ into a valid distance metric $d(X, Y)$ satisfying non-negativity, symmetry, and triangle inequality:

1 Angular Distance:

$$d_{\rho}(X, Y) = \sqrt{\frac{1}{2} (1 - \rho(X, Y))} \in [0, 1]$$

Measures directional alignment. $d_{\rho} = 0$ when $\rho = 1$; $d_{\rho} = 1$ when $\rho = -1$.

2 Absolute Correlation Distance:

$$d_{|\rho|}(X, Y) = \sqrt{1 - |\rho(X, Y)|} \in [0, 1]$$

Measures association strength regardless of sign (useful when long/short positions offset).

Topic 3.2: Shannon Entropy & Information Content

Shannon Entropy $H[X]$

Measures the uncertainty or average information content in a random variable X with probability mass function $p(x)$:

$$H[X] = - \sum_{x \in \mathcal{X}} p(x) \log_2 p(x) = E[-\log_2 p(x)]$$

- $H[X] \geq 0$ (non-negative).
- Maximized when X is uniformly distributed.
- Zero when X is deterministic.

Joint & Conditional Entropy

- **Joint Entropy** $H[X, Y]$: Uncertainty of two variables together:

$$H[X, Y] = - \sum_x \sum_y p(x, y) \log_2 p(x, y)$$

- **Conditional Entropy** $H[X|Y]$: Uncertainty remaining in X given Y :

$$H[X|Y] = H[X, Y] - H[Y]$$

Topic 3.3: Divergence & Mutual Information

Kullback-Leibler (KL) Divergence $D_{KL}[p \parallel q]$

Measures the relative entropy or information lost when approximating true distribution $p(x)$ with model $q(x)$:

$$D_{KL}[p \parallel q] = \sum_{x \in \mathcal{X}} p(x) \log_2 \left(\frac{p(x)}{q(x)} \right)$$

Note: KL divergence is asymmetric ($D_{KL}[p \parallel q] \neq D_{KL}[q \parallel p]$), so it is not a metric.

Mutual Information $I[X; Y]$

Quantifies the amount of information obtained about X through observing Y :

$$I[X; Y] = D_{KL}[p(x, y) \parallel p(x)p(y)] = H[X] + H[Y] - H[X, Y] = H[X] - H[X|Y]$$

- $I[X; Y] = 0$ if and only if X and Y are independent.
- $I[X; Y] = I[Y; X] \geq 0$ (symmetric and non-negative).

Topic 3.4: Variation of Information (VI)

Why Mutual Information Needs Modification

Mutual Information $I[X; Y]$ is a similarity measure, not a metric (does not satisfy triangle inequality). To obtain a true metric space for clustering, López de Prado utilizes **Variation of Information (VI)**.

Definition of VI

$$VI[X; Y] = H[X|Y] + H[Y|X]$$

$$VI[X; Y] = H[X, Y] - I[X; Y]$$

- Measures the information *not shared* between X and Y .
- Obeys all metric properties: triangle inequality, symmetry, non-negativity.

Normalized VI ($\widetilde{VI}$)

To compare across pairs with different entropies, normalize VI to $[0, 1]$:

$$\widetilde{VI}[X; Y] = \frac{VI[X; Y]}{H[X, Y]} = 1 - \frac{I[X; Y]}{H[X, Y]}$$

$$\widetilde{VI}_{Kraskov}[X; Y] = 1 - \frac{I[X; Y]}{\max(H[X], H[Y])}$$

Topic 3.5: Optimal Discretization & Binning

Computing Entropy on Continuous Financial Returns

Calculating $H[X]$ on continuous financial data requires partitioning data into discrete bins. Too few bins smooth out information; too many bins cause sparse estimation error.

Hacine-Gharbi & Ravier Optimal Binning Formula

For N observations of continuous variable X with correlation ρ :

$$B_X = \left\lceil \sqrt{1 + 24N \left(1 - \sqrt{1 - \rho^2}\right)^{-1}} \right\rceil$$

- Dynamically adjusts bin count based on sample size N and signal strength ρ .
- Ensures robust, unbiased estimation of $H[X]$, $H[Y]$, $H[X, Y]$, and $I[X; Y]$.

Topic 3.6: Non-Linear Codependency Demonstration

Empirical Comparison: Pearson ρ vs. Normalized Mutual Information (NMI)

Consider a non-linear relationship: $Y = 100 \cdot |X| + \epsilon$, where $X \sim \mathcal{N}(0, 1)$:

Relationship Type	Pearson Correlation (ρ)	Normalized Mutual Information (NMI)
Linear ($Y = X + \epsilon$)	0.985	0.812
Non-Linear V-Shape ($Y = X + \epsilon$)	≈ 0.000 (Fails)	0.644 (Captures Signal)
Sine Wave ($Y = \sin(X) + \epsilon$)	≈ 0.051 (Fails)	0.518 (Captures Signal)
Pure Independent Noise	0.002	0.015

Key Takeaway

Pearson correlation completely fails on non-linear relationships ($\rho \approx 0$). Mutual Information and Normalized VI ($\widetilde{VI}$) reliably detect non-linear codependency in financial time series.

Summary: Key Takeaways on Distance Metrics

Core Takeaways for Quantitative Asset Managers

- ① **Convert Correlation to Metrics:** Use $d_\rho = \sqrt{\frac{1}{2}(1 - \rho)}$ or $d_{|\rho|} = \sqrt{1 - |\rho|}$ to satisfy metric space requirements for clustering algorithms.
- ② **Look Beyond Linearity:** Financial asset codependencies are frequently non-linear. Relying solely on Pearson ρ blinds models to structural relationships.
- ③ **Use Variation of Information:** $VI[X; Y] = H[X, Y] - I[X; Y]$ provides a mathematically rigorous, bounded distance metric for asset taxonomy and feature selection.
- ④ **Apply Optimal Binning:** Use Hacine-Gharbi & Ravier binning to calculate unbiased entropy and mutual information on empirical return series.

Thank You!

Questions & Discussion

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